

SOFTWARE REQUIREMENTS SPECIFICATION

FLOWSTUDIO

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the functional and non-functional requirements of the proposed system, a web-based platform designed to connect professionals across the entertainment industry.

This document is intended for developers, project stakeholders, and evaluators to understand the system's features, behavior, and constraints. It serves as a reference for system design, development, testing, and validation.

The document ensures that all stakeholders have a clear understanding of the system requirements before the development process begins.

1.2 Scope

The proposed system is a unified web platform that enables collaboration and hiring within the entertainment industry. It allows users such as actors, musicians, models, directors, producers, writers, and crew members to create profiles, showcase their portfolios, and connect with opportunities.

The system provides features such as user registration, role-based profiles, multi-format portfolio management (video, audio, image), opportunity posting, application submission, basic messaging, and search functionality.

This platform aims to streamline the fragmented hiring and collaboration process by bringing all stakeholders into a single structured system, enabling efficient talent discovery and project execution.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
MVP	Minimum Viable Product – the basic version of the system with core features
User	Any individual registered on the platform (actor, musician, etc.)
Portfolio	A collection of user work such as videos, audio, or images
Opportunity	A project, casting call, or collaboration posted by a user
Application	A request submitted by a user to participate in an opportunity
Role	The category selected by a user (e.g., actor, musician, model)
Recruiter/ Creator	A user who posts opportunities and selects applicants
Messaging System	Feature that allows users to communicate within the platform

2. Project Overview

2.1 Project Title:

Unified Entertainment Industry Collaboration Platform

2.2 Domain:

- o Web Application
- o Entertainment Industry
- o Talent Management
- o Collaboration Systems

2.3 Description:

The proposed system is a comprehensive web-based platform developed to connect and support professionals across various domains of the entertainment industry, including actors, musicians, models, directors, producers, writers, and technical crew members.

The platform aims to provide a unified digital environment where users can create role-based profiles and present their work through a structured multi-portfolio system. This system supports different types of media content such as videos (for actors), audio files (for musicians), and images (for models), allowing users to effectively showcase their skills and previous work.

In addition to portfolio management, the platform enables users to post and manage opportunities such as casting calls, project collaborations, performance gigs, and content creation projects. Each opportunity includes relevant details such as role requirements, descriptions, and expectations. Other users can explore these opportunities and apply by submitting their profiles and portfolio content.

The system also includes an application management module that allows creators or recruiters to view applicants, evaluate their profiles, and select suitable candidates. A basic messaging system is provided to facilitate communication between users for coordination and discussion.

To enhance usability, the platform offers search and filtering capabilities, allowing users to discover talent or opportunities based on roles, skills, and location. This helps in improving visibility and accessibility within the industry.

Overall, the system is designed to address the existing challenges of fragmented communication, lack of standardized portfolios, and unstructured hiring processes in the entertainment sector. By integrating collaboration, hiring, and portfolio showcasing into a single platform, the system aims to create a more efficient, transparent, and accessible ecosystem for creative professionals.

3. Business Background

The entertainment industry largely relies on informal and fragmented methods for hiring, collaboration, and talent discovery. Traditional approaches such as personal networks, social media outreach, and manual coordination are often inefficient, unstructured, and difficult to manage at scale.

Many professionals, especially emerging artists, struggle to showcase their work effectively due to the absence of a centralized platform that supports multiple portfolio formats. Similarly, recruiters such as directors and producers face challenges in finding suitable talent quickly, as data is scattered across different sources.

Pain Points:

- **Lack of Centralized Platform:**
No single system exists that connects all roles (actors, musicians, models, etc.) in one place.
- **Inefficient Hiring Process:**
Talent discovery is time-consuming and heavily dependent on personal contacts or external platforms.
- **Poor Portfolio Management:**
Existing platforms do not support diverse media formats in a structured way (video, audio, images together).
- **Unorganized Data Handling:**
Information about users, projects, and applications is scattered and difficult to track.
- **Limited Communication Channels:**
Lack of direct and professional communication between creators and talent.
- **Low Visibility for New Talent:**
Beginners face difficulty getting exposure without strong industry connections.
- **Manual and Error-Prone Processes:**
Handling applications, shortlisting candidates, and managing collaborations manually leads to errors and inefficiencies.

This system is proposed to overcome these challenges by introducing a structured, digital, and scalable solution that streamlines collaboration and talent management within the entertainment ecosystem.

4. User Roles

The system supports multiple types of users based on their roles within the entertainment industry. Each user can register on the platform and select their role, which defines their profile type and platform interactions.

1. Actor

- Creates a profile with bio, skills, and experience
- Uploads video-based portfolio (e.g., audition clips, showreels)
- Searches and applies for casting calls and acting opportunities

2. Musician

- Creates a profile showcasing musical skills and experience
- Uploads audio portfolio (e.g., songs, compositions)
- Applies for music collaborations, performances, and projects

3. Model

- Builds a visual portfolio with images and photoshoots
- Highlights modeling experience and skills
- Applies for modeling gigs and brand collaborations

4. Director

- Posts projects and casting opportunities
- Defines role requirements and project details
- Reviews applications and selects suitable candidates

5. Producer

- Publishes opportunities such as films, events, or collaborations
- Manages applications and shortlists talent
- Coordinates with selected candidates

6. Script Writer

- Creates a professional profile showcasing writing skills and experience
- Applies to relevant projects requiring scripts or creative input
- Collaborates with directors and producers

7. Crew (Editors, Photographers, Designers, etc.)

- Showcases technical skills and previous work
- Applies for project-based roles
- Collaborates with other professionals on projects

Note:

All users can:

- Create and manage profiles
- Upload and maintain portfolios
- Search for opportunities or talent
- Apply or post opportunities (depending on role)
- Communicate using the messaging system

5. Functional Requirements

User Management

- **FR-1:** The system shall allow users to register using basic details (name, email, password).
- **FR-2:** The system shall allow users to log in and log out securely.
- **FR-3:** The system shall allow users to select a role (Actor, Musician, Model, Director, Producer, Writer, Crew) during registration.
- **FR-4:** The system shall allow users to create, view, and edit their profiles.

Profile & Portfolio System

- **FR-5:** The system shall allow users to add bio, skills, and experience to their profile.
- **FR-6:** The system shall allow users to upload portfolio content (video, audio, images).
- **FR-7:** The system shall categorize portfolio content based on type (video/audio/image).
- **FR-8:** The system shall allow users to view their own and others' profiles and portfolios.

Opportunity Posting System

- **FR-9:** The system shall allow authorized users (Directors/Producers) to post opportunities.
- **FR-10:** The system shall allow users to add details such as title, description, type, and requirements for opportunities.
- **FR-11:** The system shall allow users to view all available opportunities.

Application System

- **FR-12:** The system shall allow users to apply for opportunities.
- **FR-13:** The system shall allow users to submit their profile/portfolio during application.
- **FR-14:** The system shall allow tracking of application status (Applied, Selected, Rejected).

Application Management

- **FR-15:** The system shall allow opportunity creators to view all applicants.
- **FR-16:** The system shall allow creators to shortlist or select candidates.
- **FR-17:** The system shall allow updating application status.

Messaging System

- **FR-18:** The system shall allow users to send and receive messages.
- **FR-19:** The system shall support one-to-one communication between users.
- **FR-20:** The system shall store message history for future reference.

Search & Discovery System

- **FR-21:** The system shall allow users to search for other users and opportunities.
- **FR-22:** The system shall allow filtering based on roles, skills, and location.
- **FR-23:** The system shall display relevant search results based on user input.

6. Non-Functional Requirements

6.1 Performance

- The system shall respond to user requests within **3 seconds** under normal load conditions.
- The system shall support multiple users accessing the platform simultaneously without significant performance degradation.
- Media uploads (video/audio/images) should be processed efficiently with minimal delay.

6.2 Security

- The system shall implement **user authentication and authorization** mechanisms.
- User passwords shall be stored in **an encrypted format**.
- The system shall prevent unauthorized access to user data and private messages.
- Secure communication protocols (e.g., HTTPS) shall be used to protect data transmission.

6.3 Scalability

- The system shall be designed to support **increasing numbers of users and data**.
- The system should be deployable on **cloud platforms** (e.g., AWS) for future scaling.
- The architecture shall allow easy addition of new features and modules.

6.4 Usability

- The system shall provide a **user-friendly interface** that is easy to navigate.
- The platform shall support **clear and intuitive workflows** for profile creation, application, and communication.
- The system shall be accessible on modern web browsers.

6.5 Reliability

- The system shall ensure **high availability** with minimal downtime.
- The system shall handle errors gracefully and provide appropriate feedback to users.
- Data shall be stored reliably to prevent loss of user information and portfolio content.

6.6 Maintainability

- The system shall follow a **modular architecture** to allow easy maintenance and updates.
- Code shall be written following standard practices to ensure readability and reusability.
- Bugs and issues shall be easy to identify and fix.

6.7 Compatibility

- The system shall be compatible with **major web browsers** (Chrome, Edge, Firefox).
- The platform shall support different screen sizes (basic responsiveness).

6.8 Data Management

- The system shall store and manage user data, portfolios, and applications efficiently.
- Backup mechanisms should be in place to prevent data loss.

7. Assumptions

- Users have access to a **stable internet connection** to use the platform.
- Users possess **basic knowledge of web applications** and can navigate the system.
- Users will provide **accurate and genuine information** while creating profiles and applying for opportunities.
- The system will be accessed through **modern web browsers** such as Chrome, Edge, or Firefox.
- Media files (videos, audio, images) uploaded by users will be in **supported formats and reasonable sizes**.
- The platform will initially handle a **limited number of users** as part of the MVP.
- External services (e.g., cloud storage for media files) will be **available and function properly**.
- Users will follow basic **ethical and professional conduct** while interacting on the platform.

8. Constraints

8.1 Time Constraints

- The development of the system is limited to a duration of **2–3 months**.
- Due to limited time, only **MVP features** will be implemented.
- Advanced features (e.g., AI matching, real-time systems) are excluded in the initial version.

8.2 Technology Constraints

- The backend must be developed using **Java and Spring Boot**.
- The frontend will be implemented using **React**.
- The system will use a **relational database** (e.g., MySQL).
- Media storage will depend on **external services** (e.g., cloud storage like AWS S3 or Cloudinary).

8.3 Resource Constraints

- The project will be developed by a **limited team (students)** with restricted resources.
- Budget constraints may limit the use of **paid tools or services**.

8.4 System Constraints

- The system will initially support **basic messaging only** (no advanced real-time chat).
- File upload size and formats will be **restricted** for performance reasons.
- The platform will be designed for **web use only** (no mobile app in MVP).

8.5 Operational Constraints

- The system requires **continuous internet connectivity** for proper functioning.
- Users must use **supported browsers** for full compatibility.

9. Use Case Summary

UC-1: User Registration & Login

Field	Description
Use Case ID	UC-1
Use Case Name	User Registration & Login
Actors	All Users
Description	Allows users to register and log into the system
Preconditions	User must have a valid email
Postconditions	User account is created / user is logged in
Basic Flow	1. User enters details → 2. System validates → 3. Account created / login successful
Alternate Flow	Invalid credentials → error message

UC-2: Profile & Portfolio Management

Field	Description
Use Case ID	UC-2
Use Case Name	Profile & Portfolio Management
Actors	All Users
Description	Users create and manage their profile and portfolio
Preconditions	User must be logged in
Postconditions	Profile and portfolio updated
Basic Flow	1. User enters bio, skills → 2. Uploads media → 3. Saves profile
Alternate Flow	Invalid file format → upload rejected

UC-3: Opportunity Posting

Field	Description
Use Case ID	UC-3
Use Case Name	Opportunity Posting
Actors	Director, Producer
Description	Allows users to post casting calls or projects
Preconditions	User must be logged in
Postconditions	Opportunity is visible to other users
Basic Flow	1. Enter details → 2. Submit → 3. Opportunity published
Alternate Flow	Missing fields → error message

UC-4: Opportunity Browsing & Search

Field	Description
Use Case ID	UC-4
Use Case Name	Opportunity Browsing & Search
Actors	All Users
Description	Users search and view opportunities
Preconditions	User must be logged in
Postconditions	Relevant opportunities displayed
Basic Flow	1. Enter filters → 2. System displays results
Alternate Flow	No results → message displayed

UC-5: Application Submission

Field	Description
Use Case ID	UC-5
Use Case Name	Application Submission
Actors	Actor, Musician, Model, Writer, Crew
Description	Users apply to opportunities
Preconditions	User must be logged in
Postconditions	Application submitted successfully
Basic Flow	1. Select opportunity → 2. Click apply → 3. Submit profile
Alternate Flow	Already applied → notification shown

UC-6: Application Management

Field	Description
Use Case ID	UC-6
Use Case Name	Application Management
Actors	Director, Producer
Description	Manage and review applications
Preconditions	Opportunity must exist
Postconditions	Candidates shortlisted/selected
Basic Flow	1. View applicants → 2. Select/reject → 3. Update status
Alternate Flow	No applicants → empty list

UC-7: Messaging / Communication

Field	Description
Use Case ID	UC-7
Use Case Name	Messaging System
Actors	All Users
Description	Users communicate with each other
Preconditions	Users must be logged in
Postconditions	Messages sent and stored
Basic Flow	1. Open chat → 2. Send message → 3. Receiver gets message
Alternate Flow	Network issue → message not sent

10. Future Enhancements

The current system is designed as a Minimum Viable Product (MVP). In future versions, the platform can be enhanced with additional features to improve functionality, scalability, and user experience.

Planned Enhancements:

- **Mobile Application:**
Development of a dedicated mobile app (Android/iOS) to increase accessibility and user engagement.
- **AI-Based Recommendations:**
Implementation of intelligent algorithms to suggest:
 - Suitable candidates for opportunities
 - Relevant projects for users
based on skills, experience, and user activity.
- **Payment Integration:**
Integration of secure payment gateways to support:
 - Paid collaborations
 - Hiring transactions
 - Premium features or subscriptions
- **Advanced Real-Time Messaging:**
Upgrade messaging system to include real-time chat, notifications, and media sharing.
- **Rating and Review System:**
Allow users to rate and review each other after project completion to build trust and credibility.
- **Verification System:**
Introduce verified profiles and badges to ensure authenticity of users.
- **Analytics Dashboard:**
Provide insights such as profile views, engagement metrics, and application statistics.
- **Location-Based Matching:**
Enhance search with advanced geolocation features for local hiring and collaboration.

11. Acceptance Criteria

The system shall be considered complete and acceptable when the following criteria are satisfied:

- All features must function correctly as defined in the **Functional Requirements (FRs)**.
- User authentication and authorization must be **secure and reliable**.
- The user interface (UI) must be **responsive, user-friendly, and easy to navigate**.
- The system shall successfully execute all defined **test cases without critical errors or failures**.
- The system must meet the defined **performance benchmarks** (e.g., acceptable response time).
- All modules (user management, portfolio, opportunities, applications, messaging, and search) must work **consistently and without integration issues**.
- Data handling (user data, portfolio content, applications) must be **accurate and reliable**.
- The system shall handle **invalid inputs and errors gracefully** with proper messages.
- All features must pass their respective acceptance tests to ensure they meet **functional expectations and provide a consistent user experience**.